

## Semantic Tableaux for PL

In a tableau, every wff is prefixed with a sign,  $+$ , or  $-$ , according to the intention of finding assignments that make the wff true, or false (respectively).

The definitions and results in this lecture are taken from the book "Proof Theory and Automated Deduction" by Jean Goubault-Larrecq and I. Mackie.

### 1 Rules for semantic tableaux

Formally, we have two sets of rules in ST, based on which we obtain the tableau expansion of any wff.

#### 1.1 $\alpha$ -rules (the rules for branching of paths)

| $\alpha$                  | $\alpha_1$ | $\alpha_2$ |
|---------------------------|------------|------------|
| $+ \phi \wedge \psi$      | $+\phi$    | $+\psi$    |
| $- \phi \vee \psi$        | $-\phi$    | $-\psi$    |
| $- \phi \rightarrow \psi$ | $+\phi$    | $-\psi$    |
| $+\neg\phi$               | $-\phi$    |            |
| $-\neg\phi$               | $+\phi$    |            |

## 1.2 $\beta$ -rules (the rules for prolongation of paths)

| $\beta$                   | $\beta_1$ | $\beta_2$ |
|---------------------------|-----------|-----------|
| $- \phi \wedge \psi$      | $-\phi$   | $-\psi$   |
| $+ \phi \vee \psi$        | $+\phi$   | $+\psi$   |
| $+ \phi \rightarrow \psi$ | $-\phi$   | $+\psi$   |

We next define a tableau, step by step.

### Definition 1.1.

1. (Paths) The set of *paths* in a wff  $\phi$  is the smallest set of sets of signed wffs such that
  - (a)  $\{+\phi\}$  is a path,
  - (b) if  $C$  is a path and  $\alpha$  is a signed wff in  $C$  of type  $\alpha$ , then  $(C \setminus \{\alpha\}) \cup \{\alpha_1\}$  and  $(C \setminus \{\alpha\}) \cup \{\alpha_2\}$  are paths, where  $\alpha_1, \alpha_2$  are defined as above, and in case of  $\neg$ , the second path is omitted,
  - (c) if  $C$  is a path and  $\beta$  is a signed wff in  $C$  of type  $\beta$ , then  $(C \setminus \{\beta\}) \cup \{\beta_1, \beta_2\}$  is a path, where  $\beta_1, \beta_2$  are defined as above.
2. An *expansion strategy*  $f$  is a function from paths  $C$  with at least one non-atomic signed wff to such a wff.
3. A path  $C$  is *closed*, if there is a wff  $+\phi'$  in  $C$  such that  $-\phi'$  is in  $C$  as well.

A *tableau* is a set of paths. A tableau is *closed*, if and only if all its fully expanded paths are closed.

We state without proof, the following fundamental results of ST.

**Theorem 1.1. (Soundness, Completeness)** Tableaux are sound and complete for any strategy, i.e. for any strategy, the tableaux expansion starting from  $+\phi$  terminates on a closed tableaux, if and only if  $\phi$  is valid.

**Corollary 1.2.** If the tableau expansion of a wff has an open (i.e. not closed) path, by Theorem 1.1 the wff cannot be valid.

**Theorem 1.3. (Termination)** Let  $\phi$  be any propositional wff. The expansion of a tableau for  $+\phi$  terminates, whatever expansion strategy we choose.

**Example 1.1.** Let  $p, q$  be propositional variables.

$$\begin{array}{rcl}
 + & \neg(p \wedge q) \rightarrow (\neg p \vee \neg q) & \\
 - & \neg(p \wedge q) & \\
 + & \neg p \vee \neg q & \\
 + & \neg p & \\
 + & \neg q & \\
 - & p & \\
 - & q & \\
 + & p \wedge q & \\
 \\ 
 + & p & | \quad + \quad q
 \end{array}$$

Every path is closed, and hence the tableau is closed. By Theorem 1.1,  $\neg(p \wedge q) \rightarrow (\neg p \vee \neg q)$  is valid.

The subject in which modifications of tableaux (to make them more efficient) and other implementable proof techniques are studied, is called *Automated Theorem Proving*.